SEEDLINE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

SeedLine product requirements and specifications

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DOCUMENT CONTROL

Document ID	SDL-PRS-001	Version	0.8
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/03-requirements.md @ 0a57933		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-24	Amish Chadha	Kit 1.8 and 1.9.1 upgrade: migrated from SDL-REQ-001 to SDL-PRS-001 (PR-001 to PR-018, old R numbers in 'Was'), traced to SDL-URS-001, tolerance and units added; R11 figures from SDL-CAL-001 v0.6 (fasteners weighed from the detailed model, mass fix F6)
0.7	2026-10-02	Amish Chadha	Figures from SDL-CAL-001 v0.5 after the decisions of 2026-10-02 (13 mm slot with liners, UV-stabilized door): R11, R16 and R17 values updated; no status changed
0.6	2026-10-02	Amish Chadha	R17: USD 219.50 accepted for the prototype (SDL-DDR-003 A3, 2026-10-02); no status changed
0.5	2026-10-01	Amish Chadha	Status from SDL-CAL-001 v0.3 on the constructable design (SDL-DDR-003); R16 and R17 reported against value-engineering targets
0.4	2026-09-25	Amish Chadha	

			Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply SDL-DDR-001 (R3 kept with skip-cell plates, R5 redefined for pelleted small seed, R6 materials, R13 marker arm, R18 bolted frame); status from SDL-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each
0.1	2026-09-24	Amish Chadha	Initial scaffold

SeedLine product requirements and specifications

What the design must do, each requirement traced to a user need in SDL-URS-001. Version 0.8 migrates the requirements of SDL-REQ-001 to this format without changing any target: each R number becomes a PR with the same target, verification and status, and the old number stays in the "Was" column. The targets are still proposals checked on paper at TRL 3 in SDL-CAL-001; none has been validated with users, and they will be revised after co-design sessions (SDL-PRB-001). Statuses follow SDL-CAL-001 v0.6. The traceability matrix is generated in SDL-RTM-001. Method: STANDARDS section 22.

The **design case** is maize at 250 mm in-row spacing on 0.75 m rows, sown 50 mm deep in a tilled loam seedbed at a walking speed of 2.9 km/h (about 0.81 m/s). **Walking-speed rule** (SDL-DDR-002 item 5): walk at about 2.9 km/h or slower with the 0.8 and 1.0 ratios, and about 2.4 km/h or slower with the 1.2 ratio, so the cells stay within the assumed 0.30 m/s fill limit. The **small-seed case** is pelleted carrot or lettuce seed (about 3.3 mm pellets) at 25 to 50 mm spacing on a prepared vegetable bed; vegetable plates are the second plate set.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Single-seed placement	In the design case the seeder shall leave misses and multiples of 5 % or less of cells each, in the terms of ISO 7256-1	5	max, each	% of cells	Analysis, cell speed against the fill limit (CAL section 3), TRL 3; sticky-belt bench test, TRL 4	At risk: cell speed 0.299 m/s at the 2.9 km/h rule against an assumed 0.30 m/s limit, no margin; misses and multiples cannot be calculated	R1
PR-002	UR-002	Performance	Spacing uniformity	The coefficient of variation of in-row spacing shall be 30 % or less in the design case	30	max	%	Analysis, Monte Carlo (CAL section 10), TRL 3; field count, TRL 5	At risk: 31 % of all spacings with the assumed miss and double rates; 9 % for single spacings	R2
PR-003	UR-003	Performance	Spacing range	In-row spacing shall be settable from 25 to 400 mm by plate cell count (skip-cell plates of one or two cells included) and, where needed, one sprocket change from the optional ratio kit	25 to 400	each target within 15 %	mm	Analysis, spacing formula (CAL section 2), TRL 3	Met on paper with the ratio kit: 22 to 589 mm nominal, every target within 11.1 %; the base seeder (15 T only) covers 26 to 471 mm, within 17.3 %	R3
PR-004	UR-004	Performance	Spacing follows travel	Seed spacing shall be set by distance travelled, not speed, with wheel slip of 8 % or less	8	max	% slip	Analysis, traction estimate (CAL section 6a), TRL 3; field test, TRL 5	Met on paper: skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %, so calibrate	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									per seedbed	
PR-005	UR-003, UR-005	Performance	Seed size range	Plates of one diameter shall meter raw seed from 3.5 to 15 mm in its largest dimension; seed under 2 mm shall be sown pelleted	3.5 to 15	cell rule 1.15 L + 0.5 by 1.05 W + 0.3	mm	Analysis, cell tolerance rule (CAL section 4), TRL 3; bench test, TRL 4	Met on paper: 3.5 to 15 mm raw seed; pelleted carrot fits; raw 1 to 2 mm seed does not	R5
PR-006	UR-005	Materials	Printable plates	Each plate shall print in 2 h or less on a 180 x 180 mm bed with no supports, in PETG (default) or ASA (strong sun), PLA for trials only, with cell size within the tolerance	2	cell size +/-0.2 mm	h	Analysis, print estimate (CAL section 4), TRL 3; measurement, TRL 4	Met on paper: 1.39 h (maize) to 1.87 h (groundnut) at 20 % infill (SDL-DEC-001 F8), small margin for thick plates; accuracy to verify at TRL 4	R6
PR-007	UR-005	Usability	Crop change	A plate shall be swapped by hand with no tools, without emptying more than the seed in the housing	2	max	min	Inspection of the model, TRL 3; timed trial, TRL 4	To verify at TRL 4: side door and hand knob in the model; one 13 mm slot takes the 6, 8 and 10 mm plates with a printed liner pair	R7
PR-008	UR-006	Performance	Sowing depth	Depth shall be adjustable from 10 to 60 mm in 10 mm steps and held on a ploughed field-crop seedbed (rough, hand-	10 to 60	+/-10	mm	Analysis, depth geometry (CAL section 9), TRL 3; field check, TRL 5	Met on paper: +/-10 mm holds for drive-wheel bumps within about +/-15 mm,	R8

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				tilled seedbeds to be reviewed with partner data)					taken as the ploughed-seedbed condition	
PR-009	UR-007	Performance	Work rate	Field capacity shall be 0.1 ha/h or more in the design case, including turns and refills	0.1	min	ha/h	Analysis, field capacity (CAL section 8), TRL 3	Met on paper: 0.142 ha/h (7.1 h/ha) at 2.9 km/h	R9
PR-010	UR-008	Usability	Push effort	The horizontal push force shall be 150 N or less in the design case	150	max	N	Analysis, planar statics with a soil model (CAL section 6), TRL 3; spring-scale test, TRL 4	At risk: 132 N horizontal pushing along the handle (206 N along it) in the design case; 179 N at best on a heavy, loose seedbed	R10
PR-011	UR-009	Physical architecture	Handling	Mass with the marker kit shall be 15 kg or less; grip height adjustable from 850 to 1,050 mm; one person lifts and turns it at a row end	15	max	kg	Analysis, mass build-up (CAL section 5), TRL 3; weighing, TRL 4	Met on paper: 14.2 kg base, 14.8 kg with the marker kit (0.18 kg margin), with every fastener weighed from the detailed model and the mass fix of SDL-DEC-001 F6	R11
PR-012	UR-010, UR-017	Usability	Hopper	The hopper shall hold 2 L or more, fill from a jug without spilling and empty in 1 min or less when changing crops	2	min	L	Inspection, model volume (CAL section 8), TRL 3; timed trial, TRL 4	Met on paper: 2.42 L	R12

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-013	UR-011	Usability	Row-spacing kit	A telescoping marker arm (decided, SDL-DDR-001 item 3) shall mark the next row at 200 to 900 mm from the current row	200 to 900	+/-25	mm	Inspection of the model, TRL 3	Met by design	R13
PR-014	UR-012	Safety	Guarding	Chain and sprocket nip points shall be covered on the side facing the operator's hands and feet, with no exposed sharp edges above the opener	covered	none exposed	n/a	Inspection of the model, TRL 3	Met by design: printed shroud with an outboard face plate over both sprockets, the chain and the tensioner's full travel	R14
PR-015	UR-013	Environment and durability	Durability	Frame, drive and opener shall last 5 seasons of about 2 ha each with only chain, brush and plate replacement; plates shall last 1 season or more	5	min	seasons	Analysis, stress checks (CAL section 7), TRL 3; wear test, TRL 4	To verify at TRL 4: stresses are low, wear life unknown	R15
PR-016	UR-014	Physical architecture	Prototype cost	Prototype parts shall be reported against the USD 400 value-engineering target (`budget_usd`)	400	value-engineering target	USD	Analysis, priced BOM (`bom/bom.csv`), TRL 3	Met on paper: USD 266.10 estimated, USD 133.90 under the target	R16
PR-017	UR-015	Physical architecture	Replication cost	The base seeder (all items except the row marker kit and the optional ratio kit) shall cost	200	value-engineering target	USD	Analysis, priced BOM, TRL 3	Over target: USD 224.10 estimated, USD 24.10 over; USD 219.50 accepted	R17

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				about USD 200 in parts, a value-engineering target					for the prototype on 2026-10-02 (SDL-DDR-003 A3), and the plate liners and UV-stabilized door added USD 4.60 after it	
PR-018	UR-016	Materials	Local build	Frame, handle and opener shall be made from common steel sections with a drill and bolts (default, SDL-DDR-001 item 8) or a welder; the drive from standard #35 roller chain or bicycle parts	common parts	n/a	n/a	Inspection of the model and build plan, TRL 3	Met by design	R18

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirements not met or at risk

No requirement is not met on paper. Three are at risk and one cost requirement is over its value-engineering target:

- **PR-001 at risk:** under the walking-speed rule (2.9 km/h) the maize cells move at 0.299 m/s, at the assumed 0.30 m/s fill limit with no margin, and the limit itself needs a bench test. Plate size is revisited only with bench data (SDL-DDR-002 item 5).
- **PR-002 at risk:** the all-spacings coefficient of variation depends on miss and double rates that need a bench test.
- **PR-010 at risk:** the design case is inside 150 N with a 12 % margin on assumed soil values; a heavy, loose seedbed needs about 179 N. Whether one operator sustains the design-case push for a planting day is not yet assessed (SDL-STM-001 row S9).
- **PR-011 met with a small margin:** the kit 1.9 model weighs every bolt, nut, washer and pin, which added about 0.6 kg to the 0.58 kg fastener estimate of SDL-CAL-001 v0.5 and took the seeder with the marker kit to 15.5 kg. Under Amish's standing instruction of 2026-10-23 the mass was brought back under 15 kg (SDL-DEC-001 F6: 3

mm drop plates and bearing plate, 20 x 2 mm hopper uprights, an aluminium marker arm and a 1.5 mm marker disc): 14.8 kg with 0.18 kg of margin.

- **PR-017 over its value-engineering target:** base parts USD 224.10 against about USD 200, USD 24.10 over. Savings worth trying are in the design decisions register (SDL-DEC-001).
- **PR-008 restated:** met on paper for ploughed field-crop seedbeds; the rigid opener follows the wheels, so clods under the drive wheel on rough, hand-tilled seedbeds will change the depth.

Assumptions

All assumptions are listed in SDL-CAL-001 section 1. The main ones are a 300 mm drive wheel over the lug tips (942.5 mm travel per turn), a 0.30 m/s cell fill limit, a Brixius rolling resistance model with cone indexes of 100 to 300 kPa, and the TRL 2 miss and double rates of 5 % and 4 %.